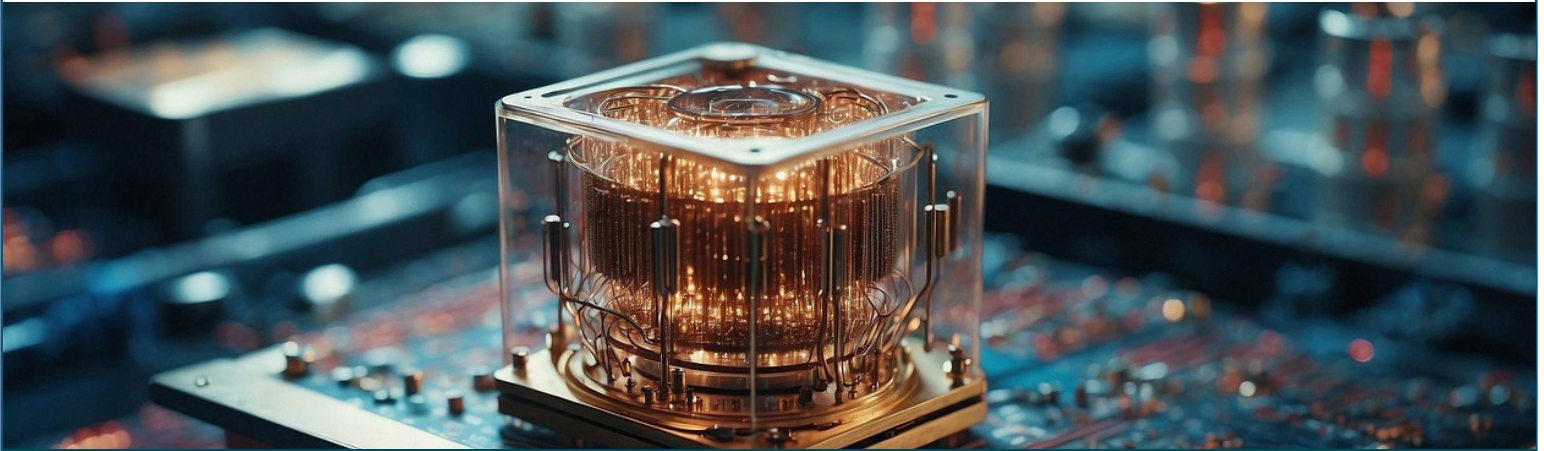




A FRAMEWORK FOR DEEPER U.S.-JAPAN QUANTUM COOPERATION

BRIEF BY HODAN OMAAR, INFORMATION TECHNOLOGY AND INNOVATION FOUNDATION

The United States and Japan are positioning their alliance for what they call a “new golden age,” built on far deeper economic and technological integration.¹ Among the advanced technologies shaping this new phase of alliance integration, quantum has emerged as a shared priority. Both governments now explicitly identify quantum information science in their national security strategies as essential to long-term economic and defense planning, with the United States’ 2025 National Security Strategy and Japan’s 2026 National Security Strategy promoting quantum as a strategic capability.² Strategic alignment alone, however, does not generate mutual benefit. How policymakers structure cooperation between the two countries’ quantum ecosystems will determine whether that alignment produces tangible gains.

The U.S. and Japanese quantum industrial bases differ in makeup, scale, and strengths in ways that create natural complementarities. If more effectively aligned, these complementarities would allow the two countries to advance quantum technologies further and faster than either could do on its own. There are three steps U.S. and Japanese policymakers could take to help translate those complementarities into practical cooperation: (1) establish a structured partnership between the U.S. National Science Foundation’s technology-commercialization directorate and Japan’s cross-ministerial innovation program to jointly commercialize quantum technologies, (2) build a bilateral quantum-infrastructure access program, modeled on the U.S. QUEST initiative, to give Japanese researchers access to U.S. quantum infrastructure, and (3) add a dedicated quantum-technology pillar to U.S.-Japan defense-industrial cooperation.

Hodan Omaar is a Senior Policy Manager focusing on AI policy in the Center for Data Innovation at the Information Technology and Innovation Foundation.

Why Quantum Technologies Are Strategically Important for the United States and Japan

The U.S.-Japan alliance has long centered on ensuring security in the Indo-Pacific and sustaining both countries' economic competitiveness. Quantum technologies are directly tied to both of those goals, shaping future military advantage and building future economic competitiveness, making quantum a critical area for bilateral cooperation.

From a security standpoint, the United States and Japan have long shared an interest in stabilizing the Indo-Pacific. Instability has intensified in recent years: North Korea is launching missiles more frequently, China is expanding military activity around Taiwan and Japan's southwest islands, and Russia is increasing naval and air operations in the western Pacific. Quantum technologies, particularly sensors, are directly disruptive to the balance of military power because they enhance the ability of any actor to detect, track, and destabilize military systems. Quantum magnetometers, for example, could help track submarines that currently operate undetected. Quantum gravimeters could reveal underground facilities or seabed movements with far greater precision. Advanced quantum inertial navigation could guide missiles or aircraft accurately, even when GPS is denied. Maintaining effective deterrence therefore depends on adopting and developing these capabilities ahead of adversaries.

From a competitiveness standpoint, quantum technologies could transform economic and industrial strength for both the United States and Japan. In the near term, quantum and quantum-inspired systems could create new sources of growth and efficiency by solving complex problems like factory scheduling, supply chain routing, and advanced portfolio optimization for finance and logistics. Over the long term, leading in foundational quantum hardware, software, and applications will strengthen national competitiveness because these

capabilities will determine who captures the next wave of high-value industries. Quantum computing, for example, could accelerate the discovery of new drugs, chemicals, and advanced materials, enabling the creation of entire sectors that do not exist today. Countries that build and control these capabilities will ensure greater productivity, higher-value manufacturing, and more resilient economic power, rather than relying on others for the technologies that underpin future growth.

Complementarities in the U.S. and Japanese Quantum Industrial Bases

While both countries recognize the strategic and economic value of innovation in quantum technologies, the United States and Japan have quantum ecosystems shaped by different industrial foundations. In the United States, progress is propelled by a dynamic venture-capital culture willing to fund high-risk, high-reward research. That environment has produced a large, diverse landscape with hundreds of firms, combining major integrated tech companies like Google and IBM with specialized, well-funded start-ups such as IonQ and Rigetti. Japan's ecosystem looks markedly different. It is anchored almost entirely by long-established industrial conglomerates—e.g., NTT, Toshiba, and Fujitsu—whose internal R&D labs direct most of the country's private quantum investment.³ With little start-up activity and few new entrants, Japan's environment is less entrepreneurial and more incumbent-driven than that of the United States.

The countries also diverge in their technological focus. The United States concentrates on core quantum-computing hardware, directing much of its effort toward developing fundamental qubit modalities—superconducting circuits, trapped ions, and other architectures that define the frontier of computation. Japan's strengths sit further up the stack. Its comparative advantage lies in the industrial implementation layer: the software, algorithms,

hybrid control systems, and precision-manufactured components that make quantum systems functional and dependable in real-world settings. This emphasis is visible in how Japan's major firms invest. Fujitsu, for example, works across the entire quantum stack, but a large share of its intellectual property and applied R&D focuses on integration-layer technologies: algorithms tailored to industrial use cases, hybrid quantum-classical control systems, and the manufacturing processes needed to link quantum processors with existing IT infrastructure.⁴ Its portfolio illustrates a broader national orientation in Japan toward the parts of the quantum stack that turn hardware into usable systems, rather than concentrating primarily on the core qubit-level technologies themselves, as in the United States. This is consistent with Japan's broader industrial tradition, evident in sectors from semiconductors to robotics, where competitive strength has often been found in turning early-stage technological advances into robust, production-ready systems that anchor entire industrial ecosystems.

Because the two ecosystems specialize in different stages of the quantum stack, cooperation carries clear commercial advantages. U.S. firms are producing frontier hardware but often face a long, uncertain path from laboratory breakthrough to industrial deployment. Japan's industrial base, by contrast, is structured to absorb, refine, and scale complex technologies within demanding manufacturing and enterprise environments. Linking U.S. hardware innovation with Japan's system-integration strengths can help both sides shorten the path from prototype to production, reduce commercialization risk, and accelerate revenue generation. Rather than duplicating capabilities, each country can fill a gap in the other's value chain, creating a pipeline that moves quantum technologies from scientific proof-of-concept to durable industrial application more efficiently than either ecosystem could do alone.

How to Accelerate U.S.-Japan Collaboration on Quantum Technologies

Before turning to the question of how to accelerate U.S.-Japan collaboration on quantum technologies, it is useful to take stock of the state of the alliance today and how quantum cooperation has shifted under the Trump administration and Japan's new prime minister, Sanae Takaichi. Previously under President Joe Biden and Prime Minister Fumio Kishida, quantum cooperation occurred within a broader security-focused agenda, alongside missile defense, space, cyber, and other emerging technologies as part of over 70 agreements aimed at strengthening Indo-Pacific deterrence and defense industrial coordination. In that context, quantum functioned as a supporting technology pillar within an overarching security and geopolitical strategy.

Under President Donald Trump and Prime Minister Takaichi, industrial and technological competitiveness have moved to the forefront of the bilateral agenda. Traditional defense cooperation will likely continue through established channels, but the political priority has moved toward economic arrangements. Initiatives such as the Framework for Securing the Supply of Critical Minerals and Rare Earths through Mining and Processing and the Technology Prosperity Deal memorandum of cooperation, which were both ratified in October 2025 during President Trump's visit to Tokyo, place quantum within a much more explicit effort to secure supply chains, expand shared production capacity, and shape the standards for next-generation technologies.

An alliance focused on industrial and technological competitiveness presents both risks and opportunities. To ensure that U.S.-Japan collaboration on quantum facilitates the alliance's full spectrum of shared goals, including deepening Indo-Pacific stability and effectively strengthening both countries' industrial bases, there are several key areas for immediate action.

First, the U.S. Congress should direct and fully fund the U.S. National Science Foundation's Directorate for Technology, Innovation and Partnerships (TIP) to serve as the primary vehicle for U.S.-Japan collaboration on quantum commercialization. TIP is the only U.S. federal directorate explicitly designed to move technologies from research into deployment, requiring industry participation and focusing on fields essential to national competitiveness, including quantum. Today, most U.S.-Japan commercial quantum activity occurs only when individual firms choose to pursue it. Quantinuum is a clear example: headquartered in Colorado, it has only two strategic commercial partnerships globally, and both are in Japan, with Mitsui and SoftBank. This pattern demonstrates two key takeaways: Japan has strong industrial demand for applied quantum solutions, and U.S. firms already see the country as a viable commercialization partner. But it also illustrates the limitation of relying on firm-by-firm initiatives alone. There is currently no U.S. federal mechanism to systematically match U.S. hardware and algorithmic innovations with Japanese industrial challenges or to scale these partnerships beyond bespoke deals, but TIP could fulfill this role.

Japan's closest counterpart to TIP is the Cross-ministerial Strategic Innovation Promotion Program (SIP), which coordinates mission-oriented R&D across ministries and industry. Structured TIP-SIP collaboration would fill the current gaps and ensure that benefits flow in both directions. A joint program could take promising U.S. quantum hardware, software, and algorithms and match them with Japan's industrial testbeds—i.e., automotive, logistics, and advanced manufacturing—where early quantum tools are already being trialed. It would also help prevent asymmetric gains. For example, Denso's work with North American quantum firms on optimization has created efficiencies for Japanese manufacturing, but there is no formal channel to facilitate equivalent deployments or market returns in the United States.⁵ A coordinated TIP-SIP model would require that

jointly supported projects be demonstrated in both economies, accelerate commercial maturity on both sides, and ensure that valuable applications emerging from collaboration strengthen both countries' industrial bases rather than only one.

Second, the United States should create a structured quantum infrastructure offering for Japan. Japan needs dependable access to quantum hardware to advance its applied R&D, develop domestic software capabilities, and train the workforce that will eventually deploy and maintain these systems. Japanese institutions are already turning to U.S. companies for this purpose. The National Institute of Advanced Industrial Science and Technology, Japan's largest public research organization, has partnered with the American quantum company IonQ to use its systems for applied research and early capability development.⁶ But deals like this are narrow and difficult to scale. They depend on one-off commercial arrangements, provide access to only a single hardware approach, and cannot deliver the stable, long-term infrastructure needed to grow a national ecosystem.

A better model comes from the United States' Quantum User Expansion for Science and Technology (QUEST) program, which created a structured mechanism for giving American universities, labs, and researchers access to U.S. quantum-computing systems.⁷ A bilateral version of this initiative, coordinated with Japan, would create a dedicated channel for Japanese universities, labs, and researchers to access U.S. quantum hardware, standardized interfaces, and shared experimentation environments. This would offer a genuine win-win outcome: Japan would gain reliable, diversified access to U.S. quantum systems for R&D, software development, and talent training, while U.S. firms would gain a stable, government-backed route into a critical market.

Third, the United States and Japan should add a dedicated quantum-technology pillar to their defense-industrial cooperation. President Trump

has pushed allies, including Japan, to assume greater responsibility for regional deterrence because Washington views its traditional role as overly costly, politically constrained, and increasingly vulnerable to China's growing military capabilities. But Japan's defense-industrial base is already overextended with its domestic obligations and cannot rapidly scale the maintenance, repair, and coproduction activities the United States is now emphasizing.

To broaden Japan's role in alliance planning, secure the Indo-Pacific more effectively, and strengthen the industrial bases of both countries, the United States and Japan should add a quantum-technology track inside the two cooperation forums they already rely on. These are the U.S.-Japan Defense Industrial Cooperation, Acquisition, and Sustainment (DICAS) forum, which manages bilateral work on military repairs, maintenance, and missile production, and the Partnership for Indo-Pacific Industrial Resilience (PIPIR), which coordinates regional projects such as shared repair hubs and drone-production standards.⁸ A quantum track within DICAS and PIPIR would give both governments a formal channel to pursue concrete joint projects, such as developing quantum sensors that

improve regional awareness, building quantum-secure communication links for crisis-resilient coordination, and running shared test sites where American and Japanese engineers can perform trials of early systems together. Because DICAS and PIPIR already conduct hands-on industrial cooperation, adding quantum to these structures is a realistic option to build new defense capabilities quickly, without creating new bureaucracy or overloading Japan's already strained industrial capacity.

Taken together, these steps would give the United States and Japan a practical way to deliver on what they have already declared this new phase of the alliance should be: a partnership built on industrial strength, coordinated technological development, and long-term competitiveness. Quantum is a domain where both countries can shape the landscape, pool complementary capabilities, and build systems that underpin military resilience and economic growth alike. If Washington and Tokyo want their "new golden age" to function as a more integrated, forward-looking alliance, then using quantum to build those foundations is a critical step to ensure that this vision moves from rhetoric to reality. ~

Endnotes

- ¹ "Implementation of the Agreement toward a New Golden Age for the U.S.-Japan Alliance," White House, October 27, 2025, <https://www.whitehouse.gov/briefings-statements/2025/10/implementation-of-the-agreement-toward-a-new-golden-age-for-the-u-s-japan-alliance>.
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THE NATIONAL BUREAU of ASIAN RESEARCH

600 UNIVERSITY STREET, SUITE 1012
SEATTLE, WA 98101 • 206-632-7370

1819 L STREET NW, NINTH FLOOR
WASHINGTON, D.C. 20036 • 202-347-9767

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